

Matrix functions and stability

NP-hardness of unrestricted static output-feedback stabilization

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because this is a longstanding complexity barrier for unrestricted feedback design; broad impact connects control synthesis, optimization, and computational complexity.

Problem statement

Consider the decision language whose inputs are rational matrices $A \in \mathbb{Q}^{n \times n}$, $B \in \mathbb{Q}^{n \times m}$, and $C \in \mathbb{Q}^{p \times n}$, with positive dimensions included in the input. The answer is yes precisely when there exists an unrestricted real matrix $K \in \mathbb{R}^{m \times p}$ for which

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A + BKC).$$

Is this language NP-hard under polynomial-time many-one reductions in the binary encoding of the rational input? The requested output is a complexity theorem, not another numerical heuristic for finding K .

References and status evidence

Minyue Fu, [Two Challenging Problems in Control Theory](#), §3, pp. 3–7, defines strict Hurwitz stabilization and distinguishes it from the known NP-hard problem with entrywise bounds on K (§3.1). Gillis and Sharma, [Solving matrix nearness problems via Hamiltonian systems, matrix factorization, and optimization](#), §3.5.2, pp. 50–51 (2022 manuscript), reiterates the unresolved complexity of static output feedback. Its broader stability convention should not replace the strict inequality above. Follow-up searches for unrestricted stabilization NP-hardness found no resolving reduction; claims about arbitrary bilinear matrix inequalities or prescribed pole placement do not by themselves settle this language. No claim of NP membership is made.

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Rechecked [Fu, §3.1](#) and [Gillis–Sharma, §3.5.2](#). Searches for unrestricted static-output-feedback NP-hardness found no applicable reduction. Bounded feedback gains, prescribed poles, and general bilinear inequalities remain different decision problems; later generic hardness descriptions do not establish this one.